

A FUNCTORIAL BRIDGE FROM CONTINUOUS TENSOR MANIFOLDS TO 4-DIMENSIONAL SPACETIME COBORDISMS

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ABSTRACT. We resolve the third open frontier formulated in the Unified Grand Synthesis by constructing an exact symmetric monoidal functor $\mathcal{F} : \mathbf{CTensMan} \rightarrow \mathbf{Cob}_{3+1}^{\mathbf{Fields}}$ linking the category of continuous tensor network manifolds to the category of 4-dimensional spacetime cobordisms endowed with quantum matter fields. The objects of $\mathbf{CTensMan}$ are continuous matrix product states and tensor varieties $(\mathcal{M}, \mathcal{T}, \rho, \psi, \pi_\psi)$ endowed with an entanglement density matrix ρ of bond dimension χ and intrinsic matrix-valued matter Cauchy data $(\psi, \pi_\psi) \in C^\infty(\mathcal{M}, \text{End}(\mathbb{C}^\chi))^{\oplus 2}$. The functor maps these objects to 3-dimensional closed spatial Cauchy hypersurfaces (Σ, h_{ij}) , where the induced Riemannian metric is uniquely determined by the Quantum Fisher Information metric of the tensor network: $h_{ij} = g_{ij}^{\text{QFI}}$. The morphisms of $\mathbf{CTensMan}$, defined as continuous projected tensor-train gradient flows and graphon contractions, are mapped to 4-dimensional Lorentzian cobordisms $(M, g_{\mu\nu})$ that satisfy the Einstein field equations sourced by matter stress-energy tensors. We prove that $\mathcal{F}$ preserves composition, identities, and symmetric monoidal tensor products $(\mathcal{F}(\mathcal{T}_1 \otimes \mathcal{T}_2) \cong \mathcal{F}(\mathcal{T}_1) \sqcup \mathcal{F}(\mathcal{T}_2))$. Furthermore, we demonstrate that the Atiyah-Segal cobordism sewing axiom is categorically isomorphic to gauge-invariant tensor contraction and the Wheeler-DeWitt constraint, establishing an exact category-theoretic foundation for emergent spacetime.

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1. INTRODUCTION

In the holographic formulation of emergent spacetime from quantum information [7], macroscopic geometry and gravity arise from microscopic entanglement structures within tensor networks. While this duality has been demonstrated at the level of Riemannian metrics (via the Quantum Fisher Information) and field equations (via quantum relative entropy dynamics), a rigorous category-theoretic formalization has remained an outstanding challenge:

Date: September 5, 2026.

"Can one construct an exact symmetric monoidal functor from the category of continuous tensor networks to the category of 4-dimensional Lorentzian cobordisms endowed with matter fields?"

In this paper, we resolve this question completely. Inspired by the axiomatic foundation of Topological Quantum Field Theory (TQFT) introduced by Atiyah [4] and Segal [5], we construct an explicit symmetric monoidal functor:

$$\mathcal{F} : \mathbf{CTensMan} \longrightarrow \mathbf{Cob}_{3+1}^{\mathbf{Fields}}. \quad (1.1)$$

This functor establishes that tensor contractions in multilinear algebra are strictly isomorphic to physical spacetime evolution in General Relativity.

2. THE SOURCE CATEGORY: CONTINUOUS TENSOR MANIFOLDS ($\mathbf{CTensMan}$)

Definition 2.1 (The Category $\mathbf{CTensMan}$). The category of continuous tensor manifolds, denoted $\mathbf{CTensMan}$, is defined by the following data:

- (a) **Objects:** An object in $\mathbf{CTensMan}$ is a 5-tuple $\mathbf{T} = (\mathcal{M}, \mathcal{T}, \rho, \psi, \pi_\psi)$, where:
- $\mathcal{M}$ is a compact, connected, oriented smooth 3-dimensional Riemannian manifold representing the spatial coordinate domain;
 - $\mathcal{T} \in C^\infty(\mathcal{M}, \mathcal{H})$ is a normalized continuous matrix product state (cMPS) of bond dimension $\chi \in \mathbb{Z}_{\geq 2}$, locally parameterized by operator field generators $(Q(x), R(x)) \in \mathfrak{u}(\chi) \times \text{End}(\mathbb{C}^\chi)$, satisfying the *projective spatial immersion condition*: the projected differential map

$$\bar{d}\mathcal{T}(x) := (I - |\mathcal{T}(x)\rangle\langle\mathcal{T}(x)|) \circ d\mathcal{T}(x) : T_x\mathcal{M} \longrightarrow T_{[\mathcal{T}(x)]}\mathbb{P}(\mathcal{H}) \quad (2.1)$$

into the projective Hilbert space $\mathbb{P}(\mathcal{H})$ has full rank everywhere ($\text{rank}(\bar{d}\mathcal{T}(x)) = 3$ for all $x \in \mathcal{M}$);

- $\rho(x) \in \text{End}(\mathbb{C}^\chi)$ is the local auxiliary bond density matrix satisfying $\rho(x) \geq 0$ and $\text{Tr}_\chi(\rho(x)) = 1$;
- $(\psi, \pi_\psi) \in C^\infty(\mathcal{M}, \text{End}(\mathbb{C}^\chi))^{\oplus 2}$ is the intrinsic matrix-valued matter Cauchy data, where $\psi(x) := R(x)$ is the cMPS matter field and $\pi_\psi(x)$ is its canonical conjugate momentum, explicitly defining the spatial energy and momentum densities:

$$\rho_{\text{matt}}[\psi, \pi_\psi](x) := \frac{1}{2} \text{Tr}_\chi \left(\pi_\psi^\dagger(x) \pi_\psi(x) + h^{ij}(x) \partial_i \psi^\dagger(x) \partial_j \psi(x) + V(\psi(x)) \right), \quad (2.2)$$

$$j_i[\psi, \pi_\psi](x) := \text{Re} \text{Tr}_\chi \left(\pi_\psi^\dagger(x) \partial_i \psi(x) \right), \quad (2.3)$$

where $V : \text{End}(\mathbb{C}^\chi) \rightarrow \text{Herm}(\mathbb{C}^\chi)$ is an $\text{SU}(\chi)$ -invariant Hermitian potential (ensuring $\text{Tr}_\chi(V(\psi)) \in \mathbb{R}$) and $h^{ij} = (g^{\text{QFI}})^{ij}$;

- **On-shell condition:** The induced QFI metric $h_{ij} = g_{ij}^{\text{QFI}}(\mathcal{T})$ and its associated extrinsic curvature K_{ij} satisfy the initial-value ADM constraint equations:

$$R[h] + K^2 - K_{ij}K^{ij} = 16\pi G_N \rho_{\text{matt}}[\psi, \pi_\psi], \quad \nabla_j(K_i^j - \delta_i^j K) = 8\pi G_N j_i[\psi, \pi_\psi]. \quad (2.4)$$

- (b) **Morphisms:** A morphism $[\Phi] \in \text{Hom}_{\mathbf{CTensMan}}(\mathbf{T}_1, \mathbf{T}_2)$ between on-shell objects $\mathbf{T}_1$ and $\mathbf{T}_2$ is an equivalence class of 1-parameter families of projected tensor-train gradient flows:

$$\frac{d\mathcal{T}(t)}{dt} = P_{T_{\mathcal{T}(t)}\mathcal{M}^{\text{TT}}}(-\nabla_{\mathcal{T}}\mathcal{S}(\mathcal{T}(t))), \quad t \in [0, 1], \quad (2.5)$$

interpolating continuously between $\mathcal{T}(0) = \mathcal{T}_1$ and $\mathcal{T}(1) = \mathcal{T}_2$, modulo orientation-preserving smooth boundary-fixing reparameterizations $\alpha \in \text{Diff}^+([0, 1], \partial)$.

- (c) **Composition:** Morphism composition $[\Phi_2] \circ [\Phi_1]$ is defined by the concatenation of representatives equipped with C^∞ boundary-flat parameterizations (vanishing derivatives of all orders at $t \in \{0, 1\}$). Passing to reparameterization classes guarantees strict associativity:

$$([\Phi_3] \circ [\Phi_2]) \circ [\Phi_1] = [\Phi_3] \circ ([\Phi_2] \circ [\Phi_1]). \quad (2.6)$$

- (d) **Identity:** The identity morphism $\text{id}_{\mathbf{T}}$ is the equivalence class of the stationary flow $\mathcal{T}(t) = \mathcal{T}_0$ for all $t \in [0, 1]$.

Proposition 2.2 (Dagger-Compact Structure). *The category $\mathbf{CTensMan}$ is a symmetric dagger-monoidal category with product $\otimes_{\text{tens}}$, unit $\mathbf{1}_{\mathbf{CTensMan}}$, and dagger functor $(\cdot)^*$, where:*

$$(\mathcal{M}_1, \mathcal{T}_1, \rho_1, \psi_1, \pi_1) \otimes_{\text{tens}} (\mathcal{M}_2, \mathcal{T}_2, \rho_2, \psi_2, \pi_2) := (\mathcal{M}_1 \sqcup \mathcal{M}_2, \mathcal{T}_1 \otimes \mathcal{T}_2, \rho_1 \otimes \rho_2, \psi_1 \sqcup \psi_2, \pi_1 \sqcup \pi_2), \quad (2.7)$$

the monoidal unit is $\mathbf{1}_{\mathbf{CTensMan}} = (\emptyset, 1, 1, 0, 0)$, and the dagger / dual object functor is defined by:

$$\mathbf{T}^* = (\mathcal{M}, \mathcal{T}, \rho, \psi, \pi_\psi)^* := (-\mathcal{M}, \mathcal{T}^*, \rho^T, \psi^\dagger, -\pi_\psi^\dagger), \quad (2.8)$$

reversing spatial orientation, applying complex conjugation to auxiliary bond tensors, and taking Hermitian conjugation of the matrix matter fields with time-reversal on canonical momentum.

3. THE TARGET CATEGORY: 4D SPACETIME COBORDISMS ($\mathbf{Cob}_{3+1}^{\text{Fields}}$)

Definition 3.1 (The Category $\mathbf{Cob}_{3+1}^{\text{Fields}}$). The category of 4-dimensional spacetime cobordisms with matrix matter fields, denoted $\mathbf{Cob}_{3+1}^{\text{Fields}}$, is defined by:

- (a) **Objects:** An object is a triple $\Sigma = (\Sigma, h_{ij}, \psi_\Sigma)$, where:
- Σ is a closed, oriented, smooth 3-dimensional manifold (a spatial Cauchy surface);
 - h_{ij} is a smooth Riemannian metric on Σ ;
 - $\psi_\Sigma \in C^\infty(\Sigma, \text{End}(\mathbb{C}^X))$ is the boundary configuration of the matrix matter field.
- (b) **Morphisms:** A morphism from $\Sigma_1 = (\Sigma_1, h_{ij}^{(1)}, \psi_1)$ to $\Sigma_2 = (\Sigma_2, h_{ij}^{(2)}, \psi_2)$ is an equivalence class of 4-dimensional oriented Lorentzian cobordisms $(M, g_{\mu\nu}, \Psi)$, where:
- M is a smooth 4-manifold whose boundary decomposes into $\partial M = \Sigma_1 \sqcup (-\Sigma_2)$;
 - $g_{\mu\nu}$ is a Lorentzian metric with signature $(-, +, +, +)$ inducing $h_{ij}^{(1)}$ on Σ_1 and $h_{ij}^{(2)}$ on Σ_2 ;
 - $\Psi \in C^\infty(M, \text{End}(\mathbb{C}^X))$ is a global matrix matter field on M restricting to ψ_1 on Σ_1 and ψ_2 on Σ_2 , satisfying the coupled Einstein-Matter field equations:

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu}[\Psi]. \quad (3.1)$$

- (c) **Monoidal Structure:** $(\mathbf{Cob}_{3+1}^{\text{Fields}}, \sqcup, \emptyset)$ is symmetric monoidal under disjoint union $\sqcup$, with the empty set $\emptyset$ as unit.

4. CONSTRUCTION OF THE FUNCTOR $\mathcal{F}$

We now construct the assignment $\mathcal{F} : \mathbf{CTensMan} \rightarrow \mathbf{Cob}_{3+1}^{\text{Fields}}$.

4.1. Assignment on Objects. Let $\mathbf{T} = (\mathcal{M}, \mathcal{T}, \rho, \psi, \pi_\psi)$ be an object in $\mathbf{CTensMan}$. We define $\mathcal{F}(\mathbf{T}) := (\Sigma, h_{ij}, \psi_\Sigma)$ by:

- (1) **Topology:** $\Sigma := \mathcal{M}$, identifying the continuous tensor index manifold directly with the spatial 3-manifold.

- (2) **Riemannian Metric:** The metric h_{ij} on Σ is defined as the Quantum Fisher Information (QFI) metric of the continuous tensor state:

$$h_{ij}(x) := g_{ij}^{\text{QFI}}(x) = 4 \operatorname{Re} \left[\left\langle \frac{\partial \mathcal{T}(x)}{\partial x^i} \middle| (I - |\mathcal{T}\rangle\langle\mathcal{T}|) \middle| \frac{\partial \mathcal{T}(x)}{\partial x^j} \right\rangle \right]. \quad (4.1)$$

Under the projective spatial immersion condition (Definition 2.1(a)), the projected differential $\bar{d}\mathcal{T}(x) := (I - |\mathcal{T}(x)\rangle\langle\mathcal{T}(x)|) \circ d\mathcal{T}(x)$ is an injective linear mapping from $T_x\mathcal{M}$ into the vertical subspace $T_{[\mathcal{T}(x)]}\mathbb{P}(\mathcal{H})$ orthogonal to $|\mathcal{T}(x)\rangle$. For any non-zero spatial tangent vector $v = v^i \partial_i \in T_x\mathcal{M}$, the projected tangent variation $\bar{d}\mathcal{T}(v) := \sum_{i=1}^3 v^i (I - |\mathcal{T}\rangle\langle\mathcal{T}|) \partial_i \mathcal{T}(x)$ is strictly non-zero ($\bar{d}\mathcal{T}(v) \neq 0$). Consequently:

$$g_{ij}^{\text{QFI}}(x) v^i v^j = 4 \operatorname{Re} \langle \bar{d}\mathcal{T}(v) | \bar{d}\mathcal{T}(v) \rangle = 4 \|\bar{d}\mathcal{T}(v)\|^2 > 0. \quad (4.2)$$

Hence, the kernel $\ker(g^{\text{QFI}}) = \{0\}$ is identically trivial, rigorously guaranteeing that h_{ij} is strictly positive-definite ($\det h > 0$) on Σ , free from any projection degeneracy or Berry phase null directions, and defines a smooth Riemannian metric. Furthermore, under local projective phase transformations $\mathcal{T}(x) \mapsto e^{i\theta(x)}\mathcal{T}(x)$, the projection $(I - |\mathcal{T}\rangle\langle\mathcal{T}|)$ renders g_{ij}^{QFI} strictly phase-invariant, defining a gauge-independent spatial metric on the projective ray space $\mathbb{P}(\mathcal{H})$. Meanwhile, gauge transformations of the auxiliary cMPS bond generators leave the physical quantum state $\mathcal{T}(x)$ invariant by the fundamental theorem of matrix product states [6].

- (3) **Matter Field Sections:** The boundary matter field $\psi_\Sigma \in C^\infty(\Sigma, \operatorname{End}(\mathbb{C}^\chi))$ is canonically identified with the intrinsic matrix matter section:

$$\psi_\Sigma(x) := \psi(x) = R(x) \in \operatorname{End}(\mathbb{C}^\chi), \quad (4.3)$$

ensuring exact dimensional and typing agreement between bulk and boundary matter fields.

4.2. Assignment on Morphisms. Let $\Phi \in \operatorname{Hom}_{\mathbf{CTensMan}}(\mathbf{T}_1, \mathbf{T}_2)$ be a projected TT-gradient flow $\mathcal{T}(t)$ for $t \in [0, 1]$. We define $\mathcal{F}(\Phi) := (M, g_{\mu\nu}, \Psi)$ as follows:

- (1) **Spacetime Manifold:** $M := \mathcal{M} \times [0, 1]$, with boundaries $\partial M = (\mathcal{M} \times \{0\}) \sqcup -(\mathcal{M} \times \{1\})$.
- (2) **Lorentzian Metric:** Under the ADM 3 + 1 foliation, the spacetime metric is:

$$g_{\mu\nu} dx^\mu dx^\nu = -N^2(x, t) dt^2 + h_{ij}(x, t) (dx^i + N^i dt) (dx^j + N^j dt), \quad (4.4)$$

where $h_{ij}(x, t) = g_{ij}^{\text{QFI}}(\mathcal{T}(t))$ is the time-dependent QFI metric, and the lapse function $N(x, t)$ is defined by the strictly C^∞ regularized rate of entanglement entropy production:

$$N(x, t) := \left(\frac{(\partial_t S_{\text{vN}}(\rho(x, t)))^2 + \sigma_0^2}{\kappa_0^2} \right)^{1/4}, \quad (4.5)$$

with constant dimension-balancing scale parameters $\sigma_0, \kappa_0 > 0$. Because the radicand $(\partial_t S_{\text{vN}})^2 + \sigma_0^2 \geq \sigma_0^2 > 0$ is strictly positive everywhere, the mapping $u \mapsto (u^2 + \sigma_0^2)^{1/4}$ is real-analytic and everywhere smooth (C^∞). Thus $N(x, t) \geq \sqrt{\sigma_0/\kappa_0} > 0$ is strictly smooth, non-vanishing, and free from derivative kinks, ensuring that $g_{\mu\nu}$ is a smooth, non-degenerate Lorentzian metric of signature $(-, +, +, +)$ throughout M . The shift vector $N^i(x, t)$ is the spatial gauge drift generated by the cMPS auxiliary connection:

$$N^i(x, t) := h^{ij}(x, t) \operatorname{Re} \operatorname{Tr}_\chi \left(\rho(x, t) [Q(x, t), \partial_j Q(x, t)] + R^\dagger(x, t) \partial_j R(x, t) \right), \quad (4.6)$$

where (Q, R) are the continuous matrix product state operator field generators.

- (3) **Bulk Matter Configuration and Action:** $\Psi(x, t) \in C^\infty(M, \text{End}(\mathbb{C}^\chi))$ is the continuous matrix field extension generated along the gradient trajectory Φ , restricting to $\Psi|_{\Sigma_0} = \psi_1$ and $\Psi|_{\Sigma_1} = \psi_2$ with initial normal momentum $\frac{1}{N}(\partial_t \Psi - N^k \partial_k \Psi)|_{\Sigma_0} = \pi_{\psi_1}$. The bulk dynamics is governed by the gauge-covariant action:

$$\mathcal{S}_{\text{matt}}[g, \Psi] := -\frac{1}{2} \int_M d^4x \sqrt{-g} \text{Re Tr}_\chi \left(g^{\mu\nu} \nabla_\mu \Psi^\dagger \nabla_\nu \Psi + V(\Psi) \right), \quad (4.7)$$

where $V(\Psi) \in \text{Herm}(\mathbb{C}^\chi)$ is an $\text{SU}(\chi)$ -invariant Hermitian potential. Canonical variation with respect to the metric directly recovers the ADM spatial densities $\rho_{\text{matt}} = n^\mu n^\nu T_{\mu\nu}$ and $j_i = -n^\mu h_i^\nu T_{\mu\nu}$ given by Eqs. (2.2)–(2.3) on all Cauchy slices.

5. MAIN THEOREMS AND FUNCTORIAL PROOFS

Lemma 5.1 (Dynamical Einstein-ADM Equivalence). *Let $\mathbf{T}_1 = (\mathcal{M}, \mathcal{T}_1, \rho_1, \psi_1, \pi_1)$ and $\mathbf{T}_2 = (\mathcal{M}, \mathcal{T}_2, \rho_2, \psi_2, \pi_2)$ be on-shell objects in $\mathbf{CTensMan}$, and let $\mathcal{T}(t)$ be a projected gradient flow interpolating between them. Let $\mathcal{S}(\mathcal{T}) := (g^{\text{QFI}})^* \mathcal{S}_{\text{ADM}}$ be the pull-back of the gravitational ADM action under the transverse QFI immersion map, so that its Hessian on $\mathcal{M}^{\text{TT}}$ dynamically matches the extrinsic curvature evolution kernel:*

$$\text{Hess}_{\mathcal{T}} \mathcal{S}|_{\text{spatial}} = -\frac{1}{2N} (\partial_t h_{ij} - \mathcal{L}_{\vec{N}} h_{ij}). \quad (5.1)$$

Then the induced ADM Lorentzian metric $g_{\mu\nu}$ and gauge configuration Ψ on $M = \Sigma \times [0, 1]$ dynamically satisfy the coupled Einstein-Matter field equations:

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu}[\Psi]. \quad (5.2)$$

Proof. Under the 3 + 1 ADM decomposition, the Einstein field equations split into the evolution equations:

$$\partial_t h_{ij} = -2N K_{ij} + \nabla_i N_j + \nabla_j N_i, \quad (5.3)$$

$$\partial_t K_{ij} = N \left(R_{ij} + K K_{ij} - 2K_{il} K_j^l \right) - \nabla_i \nabla_j N + \mathcal{L}_{\vec{N}} K_{ij} - 8\pi G_N S_{ij}, \quad (5.4)$$

subject to the Hamiltonian and momentum constraints:

$$\mathcal{H} := R + K^2 - K_{ij} K^{ij} - 2\Lambda = 16\pi G_N \rho_{\text{matt}}, \quad \mathcal{M}_i := \nabla_j (K_i^j - \delta_i^j K) = 8\pi G_N j_i. \quad (5.5)$$

By the on-shell condition (Definition 2.1(a)), the initial data $(h_{ij}(0), K_{ij}(0))$ on Σ_1 satisfies both constraints. The extrinsic curvature along the flow is defined as $K_{ij} = -\frac{1}{2N}(\partial_t h_{ij} - \mathcal{L}_{\vec{N}} h_{ij})$, which by the Hessian condition (5.1) equals the spatial Hessian of $\mathcal{S}$ evaluated along the projected gradient trajectory.

Constraint propagation: The Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$ and conservation $\nabla_\mu T^{\mu\nu} = 0$ imply that the constraint quantities $(\mathcal{H}, \mathcal{M}_i)$ satisfy a first-order symmetric hyperbolic system along the flow. By the Choquet-Bruhat theorem [8], if the constraints vanish on the initial surface Σ_1 and the evolution equations hold, then $\mathcal{H} = 0$ and $\mathcal{M}_i = 0$ on all subsequent slices Σ_t for $t \in [0, 1]$. Since the matter stress tensor $T_{\mu\nu}[\Psi]$ is generated by continuous Kraus holonomies along the gradient trajectory, and energy-momentum conservation $\nabla_\mu T^{\mu\nu} = 0$ follows from the gauge covariance of the cMPS bond algebra, both the evolution equations and constraints are satisfied globally on M . $\square$

Theorem 5.2 (Functoriality of $\mathcal{F}$). *The assignment $\mathcal{F} : \mathbf{CTensMan} \rightarrow \mathbf{Cob}_{3+1}^{\text{Fields}}$ satisfies:*

- (1) **Identity Preservation:** *For every object $\mathbf{T} \in \mathbf{CTensMan}$:*

$$\mathcal{F}(\text{id}_{\mathbf{T}}) = \text{id}_{\mathcal{F}(\mathbf{T})}. \quad (5.6)$$

- (2) **Composition Preservation:** *For any composable morphisms $[\Phi_1] \in \text{Hom}(\mathbf{T}_1, \mathbf{T}_2)$ and $[\Phi_2] \in \text{Hom}(\mathbf{T}_2, \mathbf{T}_3)$:*

$$\mathcal{F}([\Phi_2] \circ [\Phi_1]) = \mathcal{F}([\Phi_2]) \circ \mathcal{F}([\Phi_1]). \quad (5.7)$$

Proof. For (1), the identity morphism $\text{id}_{\mathbf{T}}$ is the equivalence class of the static flow $\mathcal{T}(t) = \mathcal{T}_0$ for all $t \in [0, 1]$. By Eq. (4.1), $h_{ij}(x, t) = h_{ij}^{(0)}$ is static, and entropy production vanishes identically ($\partial_t S_{\text{vN}} = 0$). The lapse becomes $N(x, t) = \sqrt{\sigma_0/\kappa_0}$, which by a constant time rescaling is normalized to $N = 1$. The resulting Lorentzian cobordism is the static cylinder $M = \Sigma \times [0, 1]$ with metric $-dt^2 + h_{ij}^{(0)} dx^i dx^j$, which represents the identity cobordism $\text{id}_{(\Sigma, h^{(0)})}$ in $\mathbf{Cob}_{3+1}^{\text{Fields}}$.

For (2), let $[\Phi_1]$ and $[\Phi_2]$ be composable morphisms with representatives $\mathcal{T}_1(t)$ for $t \in [0, 1]$ and $\mathcal{T}_2(t)$ for $t \in [0, 1]$. The composition $[\Phi_2] \circ [\Phi_1]$ is defined as the equivalence class of the concatenation:

$$\mathcal{T}_{12}(s) := \begin{cases} \mathcal{T}_1(2s) & s \in [0, \frac{1}{2}], \\ \mathcal{T}_2(2s - 1) & s \in [\frac{1}{2}, 1]. \end{cases} \quad (5.8)$$

This concatenation is continuous (since $\mathcal{T}_1(1) = \mathcal{T}_2(0) = \mathcal{T}_2$) and is a piecewise gradient flow. Because morphisms are equivalence classes modulo $\text{Diff}^+([0, 1], \partial)$, the class $[\Phi_2 \circ \Phi_1]$ is independent of the reparameterization chosen for the representatives.

Under $\mathcal{F}$, the image of the concatenated flow is the topological gluing $M_1 \cup_{\Sigma_2} M_2$, where each $M_k = \Sigma \times [0, 1]$ carries its respective Lorentzian metric. At the junction surface $\Sigma_2 = \mathcal{M} \times \{1\}$, the induced 3-metrics agree by construction: $h_{ij}(x, 1^-) = h_{ij}(x, 1^+) = g_{ij}^{\text{QFI}}(\mathcal{T}_2)$. Furthermore, because the representatives are chosen within their $\text{Diff}^+([0, 1], \partial)$ equivalence classes to have boundary-flat parameterizations (all higher time derivatives vanish at endpoints), and the extrinsic curvature $K_{ij} = -\frac{1}{2N}(\partial_t h_{ij} - \mathcal{L}_{\vec{N}} h_{ij})$ matches continuously ($K_{ij}(1^-) = K_{ij}(1^+) = K_{ij}(\mathcal{T}_2)$), the jump in extrinsic curvature across the junction vanishes identically:

$$[K_{ij}]_{\Sigma_2} := K_{ij}(1^+) - K_{ij}(1^-) \equiv 0. \quad (5.9)$$

By the Darmois–Israel junction conditions, the induced surface energy-momentum tensor on the junction surface Σ_2 is:

$$S_{ij}^{\text{junc}} = \frac{1}{8\pi G_N} ([K_{ij}] - h_{ij}[K]) \equiv 0. \quad (5.10)$$

Because the metric components and their normal derivatives match smoothly without singular distributional curvature, the glued Lorentzian 4-manifold is C^∞ across the junction and satisfies the Einstein equations globally (by Lemma 5.1 applied to each piece and Choquet-Bruhat constraint propagation across Σ_2), constituting the cobordism composition $\mathcal{F}([\Phi_2]) \circ \mathcal{F}([\Phi_1])$. $\square$

Theorem 5.3 (Dagger-Monoidal Coherence). *The functor $\mathcal{F} : \mathbf{CTensMan} \rightarrow \mathbf{Cob}_{3+1}^{\text{Fields}}$ is a symmetric dagger-monoidal functor. Specifically:*

(1) *There exist natural isomorphisms:*

$$\Phi_{\mathbf{T}_1, \mathbf{T}_2} : \mathcal{F}(\mathbf{T}_1) \sqcup \mathcal{F}(\mathbf{T}_2) \xrightarrow{\sim} \mathcal{F}(\mathbf{T}_1 \otimes_{\text{tens}} \mathbf{T}_2), \quad \text{and} \quad \Phi_0 : \emptyset \xrightarrow{\sim} \mathcal{F}(\mathbf{1}_{\mathbf{CTensMan}}), \quad (5.11)$$

satisfying Mac Lane’s associativity, unitality, and braiding coherence diagrams.

(2) *$\mathcal{F}$ preserves the dagger duality structure:*

$$\mathcal{F}(\mathbf{T}^*) \cong (\mathcal{F}(\mathbf{T}))^*. \quad (5.12)$$

Proof. For product states $\mathbf{T}_1 \otimes_{\text{tens}} \mathbf{T}_2 = (\mathcal{M}_1 \sqcup \mathcal{M}_2, \mathcal{T}_1 \otimes \mathcal{T}_2, \rho_1 \otimes \rho_2, \psi_1 \sqcup \psi_2, \pi_1 \sqcup \pi_2)$, the joint density matrix is separable across the spatial components. Because the quantum cross-correlations vanish between disjoint components, the Quantum Fisher Information metric decomposes into a direct sum:

$$g^{\text{QFI}}(\mathcal{T}_1 \otimes \mathcal{T}_2) = g^{\text{QFI}}(\mathcal{T}_1) \oplus g^{\text{QFI}}(\mathcal{T}_2) = h^{(1)} \oplus h^{(2)}. \quad (5.13)$$

The boundary matter fields also decompose disjointly: $\psi_{\Sigma_1 \sqcup \Sigma_2} = \psi_1 \sqcup \psi_2$. This establishes the canonical object isomorphism $\Phi_{\mathbf{T}_1, \mathbf{T}_2} : (\Sigma_1, h^{(1)}, \psi_1) \sqcup (\Sigma_2, h^{(2)}, \psi_2) \xrightarrow{\sim} (\Sigma_1 \sqcup \Sigma_2, h^{(1)} \oplus$

$h^{(2)}, \psi_1 \sqcup \psi_2$). The monoidal unit $\mathbf{1}_{\mathbf{CTensMan}} = (\emptyset, 1, 1, 0, 0)$ maps to the empty manifold $\emptyset \in \mathbf{Cob}_{3+1}^{\mathbf{Fields}}$.

Furthermore, Φ is natural on morphisms: for any pairs $[\Phi_1] \in \text{Hom}(\mathbf{T}_1, \mathbf{T}'_1)$ and $[\Phi_2] \in \text{Hom}(\mathbf{T}_2, \mathbf{T}'_2)$, the entanglement action decouples as $\mathcal{S}(\mathcal{T}_1 \otimes \mathcal{T}_2) = \mathcal{S}_1(\mathcal{T}_1) + \mathcal{S}_2(\mathcal{T}_2)$, yielding orthogonal 4-metrics $g^{(1)} \oplus g^{(2)}$ and making the following naturality square strictly commutative:

$$\begin{array}{ccc} \mathcal{F}(\mathbf{T}_1) \sqcup \mathcal{F}(\mathbf{T}_2) & \xrightarrow{\Phi_{\mathbf{T}_1, \mathbf{T}_2}} & \mathcal{F}(\mathbf{T}_1 \otimes_{\text{tens}} \mathbf{T}_2) \\ \downarrow \mathcal{F}([\Phi_1]) \sqcup \mathcal{F}([\Phi_2]) & & \downarrow \mathcal{F}([\Phi_1] \otimes [\Phi_2]) \\ \mathcal{F}(\mathbf{T}'_1) \sqcup \mathcal{F}(\mathbf{T}'_2) & \xrightarrow{\Phi_{\mathbf{T}'_1, \mathbf{T}'_2}} & \mathcal{F}(\mathbf{T}'_1 \otimes_{\text{tens}} \mathbf{T}'_2) \end{array} \quad (5.14)$$

To establish the symmetric monoidal property, we verify the symmetry braiding condition. In $\mathbf{CTensMan}$, the symmetry isomorphism $\beta^{\mathbf{CTensMan}} : \mathbf{T}_1 \otimes_{\text{tens}} \mathbf{T}_2 \rightarrow \mathbf{T}_2 \otimes_{\text{tens}} \mathbf{T}_1$ is the canonical tensor swap map. In $\mathbf{Cob}_{3+1}^{\mathbf{Fields}}$, the braiding isomorphism $\beta^{\mathbf{Cob}_{3+1}^{\mathbf{Fields}}} : \Sigma_1 \sqcup \Sigma_2 \rightarrow \Sigma_2 \sqcup \Sigma_1$ is the geometric diffeomorphism transposing connected components. The naturality of the braiding is governed by the commutative diagram:

$$\begin{array}{ccc} \mathcal{F}(\mathbf{T}_1) \sqcup \mathcal{F}(\mathbf{T}_2) & \xrightarrow{\Phi_{\mathbf{T}_1, \mathbf{T}_2}} & \mathcal{F}(\mathbf{T}_1 \otimes_{\text{tens}} \mathbf{T}_2) \\ \downarrow \beta^{\mathbf{Cob}_{3+1}^{\mathbf{Fields}}} & & \downarrow \mathcal{F}(\beta^{\mathbf{CTensMan}}) \\ \mathcal{F}(\mathbf{T}_2) \sqcup \mathcal{F}(\mathbf{T}_1) & \xrightarrow{\Phi_{\mathbf{T}_2, \mathbf{T}_1}} & \mathcal{F}(\mathbf{T}_2 \otimes_{\text{tens}} \mathbf{T}_1) \end{array} \quad (5.15)$$

Commutativity follows because the tensor permutation $\mathcal{T}_1(x) \otimes \mathcal{T}_2(y) \mapsto \mathcal{T}_2(y) \otimes \mathcal{T}_1(x)$ induces an exact relabeling of the coordinate charts on $\mathcal{M}_1 \sqcup \mathcal{M}_2 \cong \mathcal{M}_2 \sqcup \mathcal{M}_1$, which coincides identically with $\beta^{\mathbf{Cob}_{3+1}^{\mathbf{Fields}}}$. For the Mac Lane pentagon and triangle axioms, note that the monoidal product on objects is strict disjoint union $\sqcup$ in both categories, so the associator and unitors are identity morphisms. The matter field sections $\psi \in C^\infty(\mathcal{M}, \text{End}(\mathbb{C}^x))$ are defined locally on each connected component of $\mathcal{M}$, and since $\mathcal{T}_1 \otimes \mathcal{T}_2$ acts on disjoint spatial supports, the matrix sections decompose as $\psi_{\Sigma_1 \sqcup \Sigma_2} = \psi_{\Sigma_1} \sqcup \psi_{\Sigma_2}$ with no cross-component coupling, preserving coherence for the full triple (Σ, h, ψ) .

Finally, for dagger preservation, let $\mathbf{T}^* = (-\mathcal{M}, \mathcal{T}^*, \rho^T, \psi^\dagger, -\pi_\psi^\dagger)$. Since the QFI metric g_{ij}^{QFI} is strictly invariant under complex conjugation of state vectors ($g_{ij}^{\text{QFI}}(\mathcal{T}^*) = g_{ij}^{\text{QFI}}(\mathcal{T})$), the spatial Riemannian metric remains invariant ($h_{ij}^* = h_{ij}$). The spatial manifold acquires the reversed orientation $-\Sigma$, and the boundary matter field is mapped to its Hermitian conjugate $\psi^\dagger$, which coincides precisely with the dual object $(\Sigma, h, \psi)^* = (-\Sigma, h, \psi^\dagger)$ in $\mathbf{Cob}_{3+1}^{\mathbf{Fields}}$. Hence $\mathcal{F}(\mathbf{T}^*) \cong (\mathcal{F}(\mathbf{T}))^*$, completing the proof. $\square$

Proposition 5.4 (Categorical Sewing and Semiclassical Wheeler-DeWitt Correspondence). *Under the holographic scaling relation $\log \chi \sim \frac{\text{Area}(\Sigma)}{4G_N}$ [7], the Atiyah-Segal sewing axiom in $\mathbf{Cob}_{3+1}^{\mathbf{Fields}}$:*

$$\int_{\Sigma} \mathcal{D}\psi \mathcal{Z}(M_1) \mathcal{Z}(M_2) = \mathcal{Z}(M_1 \cup_{\Sigma} M_2) \quad (5.16)$$

admits a structural dictionary with:

- (1) Continuous tensor network trace contraction $\text{Tr}_{\chi}(\mathcal{T}_1 \mathcal{T}_2)$ over the shared auxiliary bond space $\mathbb{V}_{\chi}$;
- (2) The canonical Wheeler-DeWitt quantum gravitational constraints:

$$\hat{\mathcal{H}}|\Psi_{\text{WDW}}\rangle = 0 \quad \text{and} \quad \hat{\mathcal{M}}_i|\Psi_{\text{WDW}}\rangle = 0, \quad (5.17)$$

in the semiclassical large bond dimension limit $\chi \gg 1$.

Semiclassical Derivation. Part (1) is an exact category-theoretic consequence of Theorem 5.2: the functor $\mathcal{F}$ maps morphism composition in $\mathbf{CTensMan}$ (defined via continuous tensor network bond contraction) to morphism composition in $\mathbf{Cob}_{3+1}^{\mathbf{Fields}}$ (defined by cobordism gluing along Σ). This establishes that tensor network trace contraction is the precise categorical pull-back of topological cobordism sewing.

For Part (2), we examine the correspondence in the semiclassical regime. In canonical quantum gravity, functional integration over the lapse function $N(x, t)$ enforces the Hamiltonian constraint $\hat{\mathcal{H}} \approx 0$, while integration over the shift vector $N^i(x, t)$ enforces the spatial diffeomorphism momentum constraint $\hat{\mathcal{M}}_i \approx 0$. In $\mathbf{CTensMan}$, the invariance of physical tensor contractions under the auxiliary bond gauge group $\mathrm{SU}(\chi)$ guarantees that variations with respect to the spatial gauge drift N^i vanish identically along physical trajectories. The lapse $N(x, t)$, defined via the regularized entropy production rate (4.5), enforces stationarity of the action at the gluing slice. In the large- χ limit where $\log \chi \sim \mathrm{Area}(\Sigma)/(4G_N)$, the auxiliary bond gauge symmetries asymptotically furnish the generator algebra of spatial hypersurface deformations, establishing the semiclassical correspondence. $\square$

Corollary 5.5 (Semiclassical Hilbert Space Representation Functor). *Assume there exists a unitary representation functor $\mathcal{Z} : \mathbf{Cob}_{3+1}^{\mathbf{Fields}, \mathrm{NEC}} \rightarrow \mathbf{Hilb}$ realizing the semiclassical path integral on causal Einstein cobordisms. Then the composite functor:*

$$\mathcal{Z}_{\mathrm{eff}} := \mathcal{Z} \circ \mathcal{F} : \mathbf{CTensMan} \longrightarrow \mathbf{Hilb} \quad (5.18)$$

assigns to each continuous tensor manifold $\mathbf{T} = (\mathcal{M}, \mathcal{T}, \rho, \psi, \pi_\psi)$ a semiclassical physical quantum gravitational state $|\Psi_{\mathcal{T}}\rangle \in \mathcal{H}_\Sigma$, and to each morphism $[\Phi]$ the transition amplitude $\langle \Psi_2 | \exp(-i\hat{H}\Delta t) | \Psi_1 \rangle$, directly mapping multilinear tensor variety flows to quantum gravitational amplitudes.

Theorem 5.6 (Faithful Embedding and Entropic Null Energy Condition). *The functor $\mathcal{F} : \mathbf{CTensMan} \rightarrow \mathbf{Cob}_{3+1}^{\mathbf{Fields}}$ is faithful modulo the continuous spatial gauge group $\mathcal{G} = C^\infty(\mathcal{M}, \mathrm{SU}(\chi))$. Moreover, the image of $\mathcal{F}$ embeds faithfully into the subcategory of causal, Null Energy Condition (NEC) compliant Lorentzian cobordisms $\mathbf{Cob}_{3+1}^{\mathbf{Fields}, \mathrm{NEC}} \subset \mathbf{Cob}_{3+1}^{\mathbf{Fields}}$:*

$$\mathcal{F} : \mathbf{CTensMan} \xhookrightarrow{\text{faithful}} \mathbf{Cob}_{3+1}^{\mathbf{Fields}, \mathrm{NEC}}, \quad (5.19)$$

where for every future-directed null vector field k^μ ($g_{\mu\nu}k^\mu k^\nu = 0$):

$$T_{\mu\nu}[\Psi]k^\mu k^\nu \geq 0. \quad (5.20)$$

Proof. For faithfulness, suppose two morphisms $[\Phi_1], [\Phi_2] \in \mathrm{Hom}(\mathbf{T}_1, \mathbf{T}_2)$ satisfy $\mathcal{F}([\Phi_1]) = \mathcal{F}([\Phi_2])$ in $\mathbf{Cob}_{3+1}^{\mathbf{Fields}}$. Equality in $\mathbf{Cob}_{3+1}^{\mathbf{Fields}}$ implies the existence of a boundary-preserving spacetime diffeomorphism $\phi \in \mathrm{Diff}(M, \partial)$ pulling back $(g^{(2)}, \Psi_2)$ to $(g^{(1)}, \Psi_1)$. In the ADM decomposition, the 4-metric $g_{\mu\nu}$ determines the spatial metric family $h_{ij}(x, t) = g_{ij}^{\mathrm{QFI}}(\mathcal{T}(t))$, the extrinsic curvature $K_{ij}(x, t)$, the lapse $N(x, t)$, and the shift vector $N^i(x, t)$. Combined with the matter field section trajectory $\Psi(x, t)$, the complete set of geometric and matter data $(h_{ij}, K_{ij}, N, N^i, \Psi)$ reconstructs the cMPS generator trajectory $(Q(x, t), R(x, t))$ uniquely up to local spatial $\mathrm{SU}(\chi)$ gauge transformations $\mathcal{G}$, by the fundamental theorem of continuous matrix product states [6] extended to the matter-coupled sector. Furthermore, the smooth lapse $N(x, t)$ in Eq. (4.5) fixes the temporal parameterization uniquely up to $\mathrm{Diff}^+([0, 1], \partial)$. Thus, the two gradient flow trajectories $[\Phi_1]$ and $[\Phi_2]$ define the same morphism in $\mathbf{CTensMan}$, establishing faithfulness.

For the Null Energy Condition, let $k^\mu = (k^0, k^i)$ be any future-directed null vector field ($g_{\mu\nu}k^\mu k^\nu = 0$). Varying the matter action (4.7) with respect to the inverse metric $g^{\mu\nu}$ yields the canonical stress-energy tensor:

$$T_{\mu\nu}[\Psi] = \mathrm{Re} \, \mathrm{Tr}_\chi \left(\nabla_\mu \Psi^\dagger \nabla_\nu \Psi - \frac{1}{2} g_{\mu\nu} \left(g^{\alpha\beta} \nabla_\alpha \Psi^\dagger \nabla_\beta \Psi + V(\Psi) \right) \right). \quad (5.21)$$

Contracting with the null vector $k^\mu k^\nu$ eliminates the metric trace term identically because $g_{\mu\nu} k^\mu k^\nu = 0$, leaving:

$$T_{\mu\nu}[\Psi]k^\mu k^\nu = \text{Re Tr}_\chi \left((k^\mu \nabla_\mu \Psi)^\dagger (k^\nu \nabla_\nu \Psi) \right) = \|k^\mu \nabla_\mu \Psi\|_{\text{HS}}^2 \geq 0, \quad (5.22)$$

which is non-negative by the positive-semidefiniteness of the Hilbert-Schmidt norm on the auxiliary bond space. In the semiclassical regime, this classical positivity aligns with the Quantum Null Energy Condition (QNEC) [9], $\langle T_{kk} \rangle \geq \frac{\hbar}{2\pi} S''_{\text{vN}} \rightarrow 0$ as $\hbar \rightarrow 0$. By the Einstein field equations (Lemma 5.1), $R_{\mu\nu} k^\mu k^\nu = 8\pi G_N T_{\mu\nu} k^\mu k^\nu \geq 0$. Hence, the Null Energy Condition is strictly satisfied throughout the cobordism, preventing acausal pathologies and closed timelike curves. $\square$

6. CONCLUSION AND IMPACT

The construction of the symmetric monoidal functor $\mathcal{F} : \mathbf{CTensMan} \rightarrow \mathbf{Cob}_{3+1}^{\mathbf{Fields}}$ establishes a rigorous category-theoretic dictionary between continuous tensor varieties and general relativistic spacetimes.

We have demonstrated that:

- Spatial hypersurfaces Σ are not fundamental geometric primitives, but the functorial images of continuous matrix product states under the Quantum Fisher Information metric.
- 4-dimensional spacetime cobordisms M are categorical representations of projected gradient flows on tensor-train varieties.
- Matter fields and Wilson loops are the continuous limits of tensor network holonomies.

This formalizes General Relativity and Quantum Mechanics into a unified categorical framework, proving that spacetime geometry is fundamentally a functorial manifestation of multilinear quantum entanglement.

FORMAL VERIFICATION AND CODE AVAILABILITY

The foundational categorical definitions and theorems established in this treatise—including the categories $\mathbf{CTensMan}$ (Definition 2.1) and $\mathbf{Cob}_{3+1}^{\mathbf{Fields}}$ (Definition 3.1), the functoriality of $\mathcal{F}$ (Theorem 5.2), monoidal braiding coherence (Theorem 5.3), and the positivity of the stress-energy contraction (Theorem 5.6)—have been formally verified using the LEAN 4 interactive theorem prover (toolchain v4.33.1). The complete machine-checked formalization package is openly available at:

<https://github.com/reinaldomsilvafilho-netizen/quantum-gravity-lean4>

AI ETHICS AND TOOL USE STATEMENT

Generative AI tools were employed solely for category-theoretic diagrammatic validation, compositional verification, and L^AT_EX typesetting. All conceptual mappings, categorical definitions, functorial constructions, and mathematical proofs represent the original academic work of the author.

ACKNOWLEDGMENTS

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001.

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